

Micro I, class 7

We proved that the consumer weakly prefers a lump-sum tax to a per-unit tax; and the comparative-static result that $t \cdot (x^t - x^\ell) \leq 0$. Although we assumed that $t \in \mathbb{R}_+^L$, we never made use of the ‘+,’ so we can take $t \in \mathbb{R}^L$. Our argument there accidentally proved something more general, the *compensated law of demand*, namely, that, for any $(p, w) \gg 0$, any $\Delta p \in \mathbb{R}^L$ with $p + \Delta p \gg 0$

$$\Delta p \cdot [f(p + \Delta p, w + \Delta w) - f(p, w)] \leq 0,$$

where $\Delta w = \Delta p \cdot f(p, w)$. It asserts that, on average, prices and quantities move in opposite directions if price changes are compensated so that the consumer can afford the original choice at the new prices. It is *equivalent* to the comparative statics result in the tax problem.¹

We proved that the CLD implies that (for differentiable f)

$$\sum_i \sum_j \underbrace{\left[\frac{\partial f_i}{\partial p_j} + f_j \frac{\partial f_i}{\partial w} \right]}_{s_{ij}} \Delta p_i \Delta p_j \leq 0$$

for any $(p, w) \gg 0$ and any $\Delta p \in \mathbb{R}^L$ with $p + \Delta p \gg 0$. In sum, budget balance and WARP imply that the matrix whose i, j element is the Slutsky substitution term s_{ij} is a *negative semidefinite matrix*. Wow! The *weak* axiom seems rather misnamed, not being weak at all. You will show that BB and WARP together imply that f is homogenous of degree 0. The only property that BB and WARP does not imply is symmetry of the Slutsky substitution terms—except for the case of $L = 2$. For $L > 2$, to get symmetry the weak axiom must be strengthened to the *Strong Axiom of Revealed Preference* (SARP): if $x^1 \mathcal{R} x^2 \cdots \mathcal{R} x^T$ and $x^T \neq x^1$, then $\neg(x^T \mathcal{R} x^1)$.

We then turned to properties of *aggregate demand*. Suppose that there are I consumers. Let $d^i(p, w_i)$ be consumer i ’s demand at price-wealth pair (p, w_i) . Aggregate demand is

$$D(p, w_1, \dots, w_I) = \sum_i d^i(p, w_i).$$

We first asked: when can we write aggregate demand simply as a function of aggregate wealth? This is possible if and only if the family of demand functions $\{d^i\}_{i=1}^I$ takes the Gorman form, $d_i = A_i(p) + B(p)w_i$: the wealth expansion paths are *straight lines* with *common slope*. The family of demands takes this form if and only if the family of indirect utility functions $\{v_i\}_{i=1}^I$ have a representation of the form $v^i(p, w_i) = a_i(p) + b(p)w_i$. Special cases include identical homothetic preferences and preferences that are quasilinear in the same good (not necessarily identical preferences). In this case $V(p, w) = \sum_i a_i(p) + b(p)w$ generates the *aggregate* demand and the economy has a positive *representative consumer*: the economy’s aggregate demand is generated by a single preference relation when endowed with the economy’s total wealth. We then briefly discussed various notions of a *normative* representative consumer. I warned you that it can happen that a positive representative consumer exists, but that the preferences of the representative consumer can disagree with the unanimous preferences of the people in the economy.²

On Tuesday I will start the theory of the firm.

¹In the notation of the tax problem, set “ p ” equal to $p + t$, “ Δp ” equal to $-t$, and “ Δw ” equal to $-t \cdot f(p + t, w)$.

²I immodestly make reference to my own contribution to this large river of literature: “The value of information in efficient risk-sharing arrangements.” *American Economic Review* 91.3 (2001): 509-524.